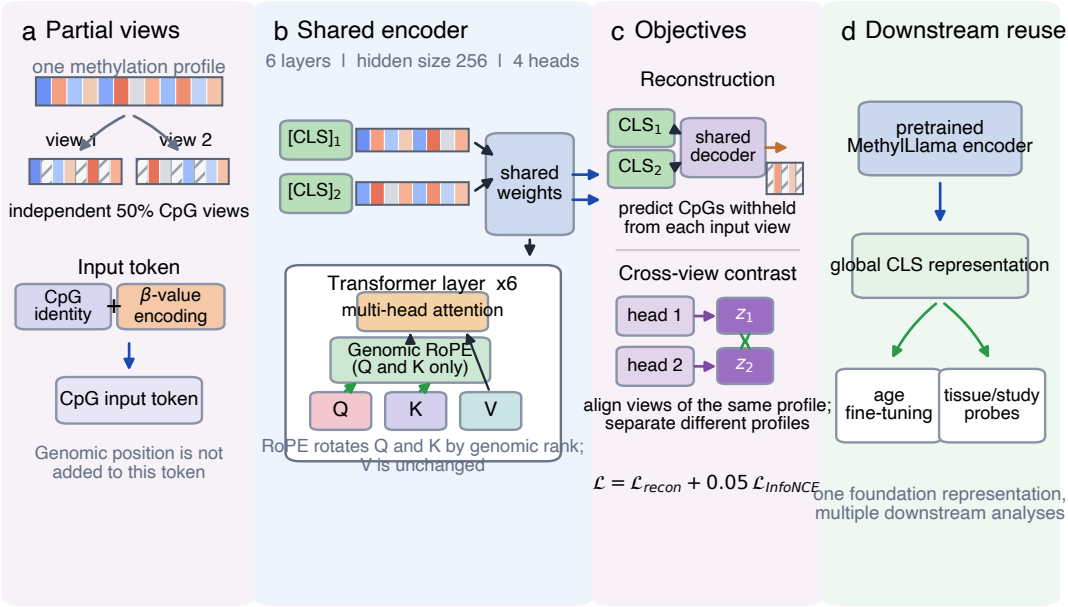
a Partial views
one methylation profile
view 1
view 2
independent 50% CpG views
Input token
CpG identity
β-value encoding
CpG input token
Genomic position is not added to this token
b Shared encoder
6 layers | hidden size 256 | 4 heads
[CLS]1
[CLS]2
shared weights
Transformer layer x6
multi-head attention
Genomic RoPE (Q and K only)
Q
K
V
RoPE rotates Q and K by genomic rank; V is unchanged
c Objectives
Reconstruction
CLS1
CLS2
shared decoder
predict CpGs withheld from each input view
Cross-view contrast
head 1
head 2
z1
z2
align views of the same profile; separate different profiles
ℒ = ℒrecon + 0.05 ℒInfoNCE
d Downstream reuse
pretrained MethylLlama encoder
global CLS representation
age fine-tuning
tissue/study probes
one foundation representation, multiple downstream analyses